



How has Physical Inactivity Changed in the American Nations From 2010 to 2020: A Decade of Lost Opportunity

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Exercise is medicine, in fact any type of physical movement provides substantial health benefits, reducing the risk of adverse health events.¹ Globally, if physical inactivity trends are not improved, ~500 million new cases of chronic disease will occur by 2030, resulting in INT\$520 billion of direct healthcare costs.² We recently published United States 2020 trends in physical inactivity according to the *American Nations* model³ developed by Colin Woodard,⁴ which discerns and maps North American regional cultures based on “first settler effects” and competing colonization patterns. The model has been applied to various historical and public policy issues, including political behavior, gender pay inequities, gun deaths and public health indices.⁵ We obtained 2010 and 2020 U.S. physical inactivity data from the County Health Rankings (CHR) program of the University of Wisconsin Population Health Institute (2010 and 2020 physical inactivity source data: Behavioral Risk Factor Surveillance System). Using zip-code data, the CHR database was linked to Nationhood Lab’s American Nations dataset.⁵ This [Figure](#) illustrates the U.S. 2010 - 2020 physical inactivity compared across the regional cultures identified by the American Nations model. Overall, physical inactivity

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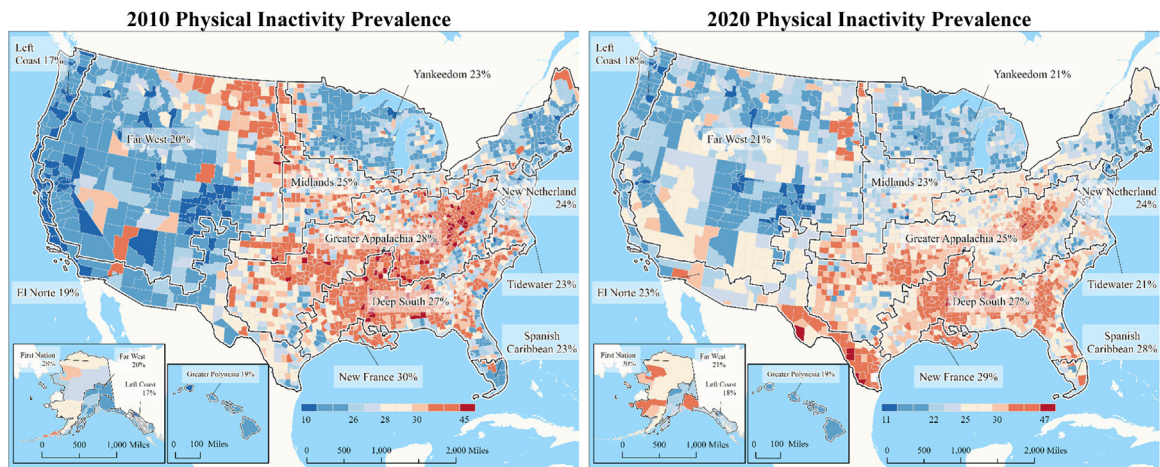


FIG. Changes in Physical Inactivity Patterns in the American Nations: 2010-2020, 2010 Physical Inactivity Prevalence, 2020 Physical Inactivity Prevalence, Physical inactivity distribution is heterogenous across the United States, Regional Cultural Differences may be a significant contributor to physical inactivity prevalence, Physical Inactivity *Hot Spots* Exist and have not sufficiently improved – new *hot spots* are emerging, The net result is overall physical inactivity prevalence is unchanged, 2010-2020 is a lost decade in terms of improving physical inactivity prevalence. (Color version of figure is available online.)

patterns have not appreciably changed: the percent of the U.S. population who were physically inactive was 24% and 23% in 2010 and 2020, respectively. On an American Nations level, several regions increased in physical inactivity prevalence (range: 0.2%-5%) while the remaining decreased (range: -0.4 to -2%). On a positive note, the Greater Appalachia region, while still considered a physical inactivity *hot spot*, has shown some improvement. Conversely, physical inactivity prevalence in El Norte and the Spanish Caribbean rose considerably. The Deep South and New France regions saw little change and remain areas of high concern. The northeast region of the Far West may be an emerging *hot spot* for physical inactivity. As stated at the outset, if the physical inactivity crisis is not addressed and rapidly reversed, the health and economic consequences will be devastating.² These findings demonstrate 2010-2020 was a lost decade in the battle against physical inactivity in the United States and should prompt significant concern amongst stakeholders from all relevant sectors (eg, government, healthcare, public health, employers, etc.). The distribution of physical inactivity in the United States is heterogeneous; established as well as emerging *hot spots* for physical inactivity pose significant challenges. The American Nations model indicates unique regional cultural differences and should be considered when crafting health promotion campaigns.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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